

Solving Word Problems Involving Volume of Cylinders

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Make sure all measurements are expressed in the same **unit of measure**.

Solving Word Problems

In solving **word problems**, determine what is asked, identify the given values, choose the appropriate formula, perform the calculations, and write the answer with the correct unit.

Examples

Example 1

A cylindrical water tank has a radius of 5 m and a height of 8 m. Find its volume.

Example 2

A cylindrical container has a diameter of 14 cm and a height of 20 cm. Determine its volume.

Example 3

A cylindrical drum can hold $1\,570\text{ cm}^3$ of oil. If its radius is 5 cm , find its height.

Practice Exercises

Practice 1

A cylindrical water tank has a radius of 5 m and a height of 8 m. Find its volume.

Practice 2

A cylindrical container has a diameter of 14 cm and a height of 20 cm. Determine its volume.

Practice 3

A silo has a radius of 6 m and a height of 15 m.
How many cubic meters of grain can it hold?

Practice 4

A cylindrical flower pot has a diameter of 24 cm and a height of 30 cm. Find its capacity.

Practice 5

A cylindrical drum can hold $1\,570\text{ cm}^3$ of oil. If its radius is 5 cm , find its height.

Practice 6

A cylindrical container has a volume of 2 464 cm³. If its height is 16 cm, find its radius. Use

$$\pi = \frac{22}{7}.$$

Activities

Activity 1

A juice can has a radius of 3 cm and a height of 15 cm. How much juice can it hold?

Activity 2

A pillar is shaped like a cylinder with radius 4 cm and height 25 cm. Find its volume.

Activity 3

A storage silo has a radius of 8 m and a height of 12 m. How many cubic meters of corn can it hold?

Activity 4

A cylindrical bucket has a diameter of 28 cm and a height of 35 cm. Find its capacity.

Activity 5

A pipe has an inside radius of 3 cm and a length of 90 cm. Assuming it is a cylinder, determine its volume.

Activity 6

A cylindrical drum can hold $2\,772\text{ cm}^3$ of liquid. If its radius is 7 cm , find its height. Use $\pi = \frac{22}{7}$.

Activity 7

A cylindrical aquarium has a radius of 28 cm and a height of 75 cm. Water costs ₱18 per 1 000 cm³. How much will it cost to completely fill the aquarium? Use $\pi = \frac{22}{7}$.

Thank you!